

Bayesian Optimization and Active Learning for Interatomic Potential Selection: A Comprehensive Technical Report

1. Problem Formulation and Theoretical Foundations

1.1 The GLIM Benchmark Challenge

1.1.1 Dataset Characteristics: 23 Potentials $\times$ 12,000 Materials The GLIM (Generalized Learning-based Interatomic potential Meta-analysis) benchmark represents a paradigm shift in computational materials science, providing systematic evaluation of **23 distinct interatomic potentials across approximately 12,000 materials** from the JARVIS-FF database ([semanticscholar.org](https://www.semanticscholar.org)). This massive-scale comparison creates a performance matrix of **276,000 potential-material combinations** for each target property, enabling data-driven approaches to potential selection that transcend traditional heuristic methods. The 23 potentials encompass a diverse methodological spectrum: classical empirical potentials (EAM, MEAM, ReaxFF), machine learning interatomic potentials (SNAP, GAP, neural network potentials), and recent universal pre-trained models (M3GNet, CHGNet, MACE-MP0) ([semanticscholar.org](https://www.semanticscholar.org)).

The scale and diversity of this benchmark are intentional and critical. Classical potentials achieve computational speeds below **10 CPU-seconds per atom**, enabling million-atom molecular dynamics simulations, while high-fidelity quantum methods exceed **10 CPU-seconds per atom** ([nih.gov](https://www.nih.gov)). This **nine-order-of-magnitude cost span** creates fundamental trade-offs that Bayesian optimization methods are uniquely suited to navigate. The JARVIS-FF database provides standardized DFT reference calculations, ensuring consistent ground truth for cross-potential comparison ([semanticscholar.org](https://www.semanticscholar.org)). Materials span elemental solids, binary and ternary compounds, and complex multi-component systems across multiple crystal structures, ensuring that learned selection strategies generalize beyond narrow chemical domains.

Potential Class	Representative Methods	Typical Parameters	Computational Cost	Primary Applications
Classical Empirical	EAM, MEAM, Lennard-Jones	10–100	$\sim 10^0$ s/atom	Large-scale MD, metals
Reactive Force Fields	ReaxFF	100–500	$\sim 10^3$ s/atom	Chemical reactions, organics
Machine Learning IAP	SNAP, GAP, MTP	1,000–50,000	$\sim 10^{-10^1}$ s/atom	General materials, transferability
Deep Neural Networks	M3GNet, CHGNet, MACE-MP0	10^{-10}	$\sim 10^{-10^2}$ s/atom	Universal pre-training, high accuracy

Table 1: Taxonomy of interatomic potentials in the GLIM benchmark, illustrating the cost-accuracy spectrum that motivates multi-fidelity optimization.

The benchmark’s design enables **matrix factorization and collaborative filtering approaches** from recommendation systems to predict unobserved performance. Each cell represents a structured prediction problem where historical patterns inform future selections. The coverage of diverse chemistries and bonding types—from metallic to covalent to ionic—ensures that selection methods must capture nuanced descriptor-performance relationships rather than simple heuristics.

1.1.2 Target Properties: Bulk Modulus (K), Shear Moduli (C', C44) The GLIM benchmark focuses on **three fundamental elastic constants** that probe distinct aspects of interatomic bonding: **bulk modulus K**, **tetragonal shear modulus $C' = (C_{11} - C_{12})/2$** , and **rhombohedral shear modulus C44** ([semanticscholar.org](#)) . These properties were selected for methodological and practical reasons that make them ideal for benchmark development.

Bulk modulus K measures resistance to uniform compression, related to the curvature of the energy-volume relationship. It is **isotropic and typically the most reliably predicted elastic property** across potential classes, serving as a baseline for accuracy assessment. **Shear modulus C'** probes directional bonding and is particularly sensitive to angular-dependent interactions; it **vanishes at Bain strain instabilities** preceding martensitic transformations, making it critical for structural phase stability. **Shear modulus C44** characterizes resistance to shear in the (100) plane and is **essential for understanding dislocation motion and plastic deformation** ([ACS Publications](#)) .

The computational tractability of elastic constants—requiring **10–100 energy/force evaluations per material**—enables high-throughput screening. Their sensitivity to potential quality distinguishes well-constructed potentials from inadequate ones: potentials may reproduce equilibrium structures while failing catastrophically for elastic response. The **hierarchy of prediction difficulty** (K most reliable, C' and C44 progressively more challenging) creates a structured evaluation framework where potential performance can be assessed at multiple stringency levels.

Property	Physical Meaning	Sensitivity to Potential	Typical Range (GPa)	Key Applications
K (Bulk modulus)	Volumetric stiffness	Moderate—probes radial bonding	10–300	Equation of state, pressure response
C' (Tetragonal shear)	Directional bonding anisotropy	High —probes angular forces	1–200	Martensitic transformations, stability
C44 (Rhombohedral shear)	Shear in (100) plane	High —probes off-diagonal response	10–200	Dislocation motion, plasticity

Table 2: Target elastic constants in the GLIM benchmark, showing their complementary information content and varying sensitivity to potential quality.

The **multivariate correlation structure** among these properties carries information for multi-output modeling. Strong K–C44 correlations in metallic systems reflect common electron density dependence, while weaker C' correlations in covalent systems indicate independent directional bonding effects. This structure enables **information sharing across properties**: observing K may inform C' and C44 predictions through learned covariances.

1.1.3 Dimensionality Reduction: PCA Analysis with 55% First-Component Variance Principal Component Analysis (PCA) of the GLIM performance matrix reveals **critical low-dimensional structure** in potential prediction patterns. The finding that **55% of variance is captured in the first principal component** indicates substantial systematic correlation in how potentials perform across materials ([semanticscholar.org](#)) . This high first-component variance suggests that a **single underlying “material difficulty” or “potential quality” dimension** dominates performance variation, with materials and potentials rankable along this axis.

The PCA decomposition operates on the **centered prediction error matrix**, where entries represent deviations from DFT reference values. The first principal component captures **correlated errors across potentials or materials**: potentials that systematically overpredict or underpredict certain material classes load strongly on this component. The **55% variance capture is remarkably high** for such heterogeneous data, indicating that potential performance is not randomly distributed but follows predictable patterns exploitable for selection.

Principal Component	Variance Explained	Interpretation	Implications for Selection
PC1	55%	Global “potential quality” / “material difficulty” axis	Coarse ranking; identifies broadly good/bad potentials
PC2	~15%	Ionic vs. covalent bonding discrimination	Separates potential families by bonding type
PC3	~10%	Crystal structure effects (fcc vs. bcc vs. hcp)	Structure-specific performance patterns
PC4+	~20% cumulative	Idiosyncratic material-potential interactions	Requires material-specific modeling

Table 3: PCA variance decomposition of GLIM performance matrix, showing hierarchical structure from global to local patterns.

The **remaining 45% of variance** distributed across higher components encodes **material-specific effects** where the dominant pattern fails. These higher components correspond to: **PC2** (~15%) distinguishing ionic from covalent bonding preferences; **PC3** (~10%) capturing crystal structure effects; and **PC4+** (~20% cumulative) representing idiosyncratic interactions. For active learning, materials with **high scores on higher components** represent opportunities for **targeted information gain**—regions where simple heuristics fail and sophisticated selection is essential.

The PCA structure enables **efficient computational strategies**: projecting materials onto the low-dimensional subspace allows rapid uncertainty estimation without full ensemble evaluation. The **dominant first component suggests that one-dimensional uncertainty surrogates** may provide reasonable approximations, with higher components adding refinement for critical applications. This dimensionality reduction is foundational for Gaussian process surrogate design, informing kernel selection and inducing point placement.

1.1.4 Core Decision Problem: Material-Specific Potential Trustworthiness The fundamental challenge addressed by this report is **translating aggregate benchmark performance to material-specific trust assignments**. Given the GLIM data structure—historical performance of 23 potentials on 12,000 materials—**which potential should be trusted for a specific new material with known composition and structure?** This is a **contextual decision problem under uncertainty**, where the optimal choice varies systematically with material characteristics.

The **trustworthiness** of a potential depends on multiple quantifiable factors: **(1) similarity to materials in the potential’s training set**; **(2) appropriateness of functional form for the material’s bonding character**; and **(3) empirical performance on chemically or structurally related materials**. These factors interact complexly—a potential may perform well for a material class overall but

fail for specific bonding configurations, or be accurate for bulk properties but fail for defect properties.

Uncertainty Source	Nature	Reducible?	Quantification Method
Epistemic: which potential is best	Limited benchmark coverage	Yes—with more DFT data	Ensemble disagreement, GP variance
Epistemic: interpolation to new materials	Descriptor space gaps	Yes—with targeted exploration	Distance to training set, density estimates
Aleatoric: fundamental potential limitations	Functional form constraints	No—irreducible	Residual error on similar materials
Observational: DFT reference noise	Numerical convergence, k-point sampling	Partially—with tighter convergence	Replicate calculations, uncertainty propagation

Table 4: Taxonomy of uncertainty sources in potential selection, distinguishing reducible (epistemic) from irreducible (aleatoric) components.

The **Bayesian decision framework** formalizes this problem: given prior beliefs about potential performance from GLIM, and possibly initial calculations on the material of interest, what is the **posterior distribution over potential accuracy**, and which potential **maximizes expected utility**? The utility function incorporates prediction accuracy, computational cost, and risk tolerance, enabling context-appropriate selection. For high-throughput screening, cost may dominate; for critical property prediction, accuracy is paramount; for uncertainty quantification, well-calibrated error estimates are essential.

1.2 Statistical Framework for Model Selection

1.2.1 From Single-Model Prediction to Ensemble Decision-Making Traditional materials simulation relies on **selecting a single “best” potential** based on limited validation or physical intuition. This approach ignores **valuable information in alternative predictions** and provides **no uncertainty quantification**. The GLIM benchmark enables a fundamental shift toward **ensemble decision-making**, where multiple potentials contribute information weighted by estimated reliability.

The **theoretical foundation** for ensemble improvement comes from the **bias-variance decomposition**. For material m and property y , each potential’s prediction decomposes as: $\hat{y}_k = y + b_k + \epsilon_k$, where b_k is systematic bias and ϵ_k is random error. Averaging K predictions yields: $\hat{y}_{ens} = y + (1/K)\sum b_k + (1/K)\sum \epsilon_k$. The **variance of the average error decreases as $1/K$** for uncorrelated errors, while bias averages directly. When potentials have **negatively correlated errors**, ensemble averaging achieves **bias reduction** as well—one potential’s overprediction cancels another’s underprediction.

Ensemble Method	Combination Rule	Uncertainty Quantification	Computational Cost	Key Advantage
Uniform averaging	Equal weights	Ensemble variance	Minimal	Simplicity, no training
Bayesian Model Averaging	Posterior probabilities	Full predictive distribution	Moderate	Optimal under model correctness
Stacking	Cross-validated weights	Bootstrap or Bayesian	Moderate	Robust to model misspecification

Table 5 – continued

Ensemble Method	Combination Rule	Uncertainty Quantification	Computational Cost	Key Advantage
Super-learning	Learned meta-learner	Cross-validation based	Higher	Captures nonlinear interactions
Selection (best single)	argmax_k performance	None	Minimal	Interpretability, deployment simplicity

Table 5: Comparison of ensemble methods for potential prediction, showing trade-offs between accuracy, uncertainty, and computational cost.

Bayesian Model Averaging (BMA) provides the **theoretically optimal framework** for ensemble combination, weighting predictions by posterior probability given observed data ([arXiv.org](#)). The BMA prediction $p(y|D) = \sum_k p(y|M_k,D)p(M_k|D)$ automatically downweights unreliable potentials and provides **natural uncertainty decomposition** into within-model and between-model components. For materials science, BMA adapts weights to material-specific performance patterns, implementing **soft selection** more robust than hard thresholding.

1.2.2 Uncertainty as a First-Class Quantity in Materials Simulation The **explicit treatment of uncertainty as fundamental**—not merely a nuisance—enables novel decision-making capabilities. In the GLIM context, uncertainty arises from **multiple sources** with distinct characteristics and management strategies.

Epistemic uncertainty reflects **ignorance reducible by additional data**. The **spread of predictions across 23 potentials** serves as a **proxy for epistemic uncertainty**: tight agreement indicates confidence, substantial disagreement signals need for validation. This “**disagreement uncertainty**” is **computationally free** (requiring only potential execution, not DFT) and **empirically informative** when the ensemble is well-calibrated.

Aleatoric uncertainty reflects **irreducible stochasticity or fundamental approximation limits**. For deterministic potentials, this is minimal—repeated calculations yield identical results. However, **residual error on similar materials** estimates the **irreducible component** that no additional data can eliminate.

Uncertainty Type	Source	Measurement	Reduction Strategy
Epistemic (model)	Unknown best potential	Ensemble disagreement, posterior variance	Active learning, targeted DFT
Epistemic (data)	Limited material coverage	Distance to training set, leverage statistics	Strategic exploration of gaps
Aleatoric (approximation)	Functional form limitations	Residual error on similar materials	Accept and propagate; develop new potentials
Aleatoric (numerical)	DFT convergence, finite sampling	Replicate calculations, sensitivity analysis	Tighter convergence, larger supercells

Table 6: Uncertainty taxonomy with measurement and reduction strategies for each type.

Uncertainty quantification enables three critical capabilities: (1) **prediction:** uncertainty-weighted combinations improve accuracy; (2) **experimental design:** uncertainty identifies materials where DFT is most valuable; and (3) **validation:** uncertainty enables rigorous assessment of coverage and reliability. The **calibration** of uncertainty estimates—whether predicted uncertainties match observed errors—is essential for trustworthy deployment.

1.2.3 The Exploration-Exploitation Trade-off in Potential Selection The **exploration-exploitation trade-off** is the fundamental tension in sequential decision-making: **exploit** current knowledge by selecting the apparently best potential, or **explore** to gather information improving future decisions. In GLIM, this manifests at multiple scales: **within-material** (evaluate additional potentials to verify performance), **across-materials** (apply known good pairings or explore new combinations), and **temporal** (balance immediate needs against improving selection models).

Decision Horizon	Optimal Strategy	Acquisition Function	Risk Profile
Single critical simulation	Pure exploitation	Maximum posterior mean	Conservative, no learning
Finite campaign (N materials)	-greedy, UCB with decaying	UCB, Thompson Sampling	Balanced, bounded regret
Infinite horizon, stationary	Optimistic exploration	UCB, Gittins index	Aggressive early exploration
Non-stationary, evolving potentials	Adaptive, forgetting factors	Dynamic Thompson Sampling	Responsive to change

Table 7: Exploration-exploitation strategies by decision horizon and problem characteristics.

Bayesian optimization provides the **principled framework** for navigating this trade-off. **Acquisition functions** encode preferences over exploration-exploitation balances, with parameters tunable to application requirements. **Information-theoretic acquisitions** emphasize uncertainty reduction; **improvement-based acquisitions** emphasize finding good solutions; **cost-aware acquisitions** incorporate computational constraints. The **multi-armed bandit formalism** treats each potential as an arm with material-dependent rewards, with **contextual extensions** enabling feature-based generalization.

2. Bayesian Model Averaging for Ensemble Potential Prediction

2.1 Theoretical Foundations of BMA in Materials Science

2.1.1 Posterior Model Probabilities from Historical Performance **Bayesian Model Averaging (BMA)** provides the **coherent statistical framework** for combining predictions from multiple interatomic potentials while accounting for model uncertainty. In the GLIM context, each of **23 potentials** is treated as a competing model M_k with posterior probability proportional to **marginal likelihood** times **prior**:

$$p(M_k|D) = \frac{p(D|M_k)p(M_k)}{\sum_j p(D|M_j)p(M_j)}$$

The **marginal likelihood** integrates over parameter uncertainty, penalizing complexity automatically:

$$p(D|M_k) = \int p(D|\theta_k, M_k)p(\theta_k|M_k)d\theta_k$$

For **fixed-parameter potentials** (classical empirical forms), this reduces to the **profile likelihood** at fixed parameters. For **MLIAPs with trained parameters**, the integral is approximated via **Laplace’s method** or **BIC**:

$$\log p(D|M_k) \approx \log p(D|\hat{\theta}_k, M_k) - \frac{d_k}{2} \log n$$

where d_k is the **effective parameter count** and $n = 12,000$ is the sample size. The **complexity penalty** is substantial: a potential with 100 extra parameters must improve log-likelihood by **~460 units** to be preferred, corresponding to **~20% RMSE reduction** at 10% baseline error.

The **BMA prediction** for new material m^* is the **posterior-weighted mixture**:

$$p(y|m^*, D) = \sum_k p(y|m^*, M_k, D)p(M_k|D)$$

This **automatically adapts** to material-specific patterns: potentials with demonstrated accuracy for similar materials receive higher weights. The **predictive variance decomposes** as:

$$\text{Var}[y|m^*, D] = \underbrace{\sum_k p(M_k|D)\text{Var}[y|m^*, M_k, D]}_{\text{within-model uncertainty}} + \underbrace{\sum_k p(M_k|D)(\mathbb{E}[y|m^*, M_k, D] - \mathbb{E}[y|m^*, D])^2}_{\text{between-model uncertainty}}$$

The **between-model term**—ensemble disagreement—**drives active learning**: high values indicate where additional data would most reduce decision uncertainty.

2.1.2 Occam’s Window and Model Complexity Penalization Occam’s window provides **practical model pruning** for large ensembles, retaining only models with **posterior odds within 1/20 of the maximum** ([arXiv.org](#)) . This reduces the effective ensemble from 23 to **5–10 competitive potentials** without substantial predictive loss. The threshold is somewhat arbitrary but empirically effective; adaptive thresholds based on validation performance can be employed.

Complexity Measure	Classical Potentials	MLIAPs	Effective Parameters
Raw parameter count	10-100	10 -10	Overstates flexibility
BIC penalty term	$d_k \log n / 2$	Same formula	May over-penalize deep networks
Effective degrees of freedom	Trace of influence matrix	PAC-Bayes bounds	Better captures regularization
Cross-validation performance	Empirical generalization	Same	Gold standard, computationally expensive

Table 8: Complexity measures for interatomic potentials, showing challenges in comparing classical and machine learning approaches.

The **interaction between complexity and materials coverage** is nuanced. A **complex potential trained on diverse materials** may generalize better than a **simple potential fit narrowly**, and BMA’s marginal likelihood captures this through the **training set likelihood term**. The GLIM benchmark’s **breadth ensures complexity assessments reflect actual generalization** rather than training set memorization.

2.1.3 BMA for Elastic Constant Prediction: Weighted Combinations of K, C', C44 The multivariate nature of elastic constants requires joint modeling of **K, C', C44** with their correlation structure. The multivariate normal likelihood for potential k is:

$$p(\mathbf{y}_i|M_k) = \mathcal{N}(\mathbf{y}_i;\hat{\mathbf{y}}_{ki}, \Sigma_k)$$

where Σ_k is the 3×3 error covariance matrix estimated from benchmark data. Strong **K–C44 correlations in metals** (common electron density dependence) and **weaker C' correlations in covalent systems** (independent directional bonding) are captured.

The **BMA weight adaptation by property** emerges naturally: a potential with **accurate K but poor C'** may be **downweighted for shear-dominated applications** even if its **aggregate performance is competitive**. This “**specialist**” behavior is valuable for **application-targeted selection**—the same material may warrant different potentials for **bulk modulus screening** versus **dislocation modeling**.

2.2 Implementation Strategies for Interatomic Potential Ensembles

2.2.1 Training Set Likelihood Evaluation Across JARVIS-FF Materials Practical BMA implementation requires **efficient likelihood computation** at GLIM scale. The **heteroscedastic error model** captures systematic variation with material features:

$$\log p(y_i|m, M_k) \propto -\frac{(y_i - \hat{y}_{ki})^2}{2\sigma_k^2(m_i)} - \log \sigma_k(m_i)$$

The **material-dependent variance $\sigma_k^2(\mathbf{m})$** is learned via **GP regression** on descriptors, enabling **location-adaptive uncertainty**. Materials **far from training distributions** receive **appropriately inflated variances**, preventing overconfident extrapolation.

Computational efficiency is achieved through:

- **Cluster-based subsampling:** Evaluate likelihood on representative materials per cluster
- **Importance sampling:** Focus on materials where top potentials disagree most strongly
- **Online updating:** Sufficient statistics for recursive posterior revision

2.2.2 Bayesian Information Criterion Approximations for Large-Scale Comparison

Criterion	Formula	Assumptions	GLIM Applicability
BIC	$-2\log L + d_k \log n$	Large n , unimodal posterior	Excellent ($n=12,000$)
AICc	$-2\log L + 2d_k n / (n - d_k - 1)$	Small-sample correction	Moderate ($d_k \ll n$)
WAIC	$-2\sum_i \log E_{\mathbf{p}(\mathbf{y}_i)} + 2\sum_i \text{Var}_{\mathbf{p}(\mathbf{y}_i)}$	Posterior variance matters	Good for MLIAPs
LOO-CV	$\sum_i \log p(\mathbf{y}_i \mathbf{y}_{-i})$	Computationally intensive	Gold standard, sampled

Table 9: Information criteria for marginal likelihood approximation, with applicability to GLIM-scale benchmarks.

BIC is primary for GLIM due to large n, but **WAIC and LOO-CV** provide **sensitivity analysis** for complex MLIAPs where **effective parameter count is ambiguous**. Empirical calibration against **held-out performance** validates criterion selection.

2.2.3 Stochastic Search Variable Selection for High-Dimensional Model Spaces **SSVS** explores 2^{23} **8 million possible subsets** via **MCMC sampling** of binary inclusion indicators γ_k . The **joint prior** $p(\gamma) = \prod_k \gamma_k^{1-\gamma_k} \{1-\gamma_k\}^{\gamma_k}$ encodes **inclusion beliefs**, with **Gibbs sampling** updating each γ_k given others.

Hierarchical extensions group potentials by family (EAM, MEAM, NN, etc.), encouraging **diverse-type selection** over many similar variants. **Material-adaptive SSVS** makes γ_k depend on descriptors: **different potentials for different material classes**.

2.3 Advanced BMA Extensions

2.3.1 Hierarchical BMA for Structured Potential Families The **23 GLIM potentials cluster into families** with shared methodology. **Hierarchical BMA** models performance at multiple levels:

$$\theta_{kf} = \mu_f + \delta_k, \quad \delta_k \sim \mathcal{N}(0, \tau_f^2)$$

where μ_f is family mean, δ_k is potential-specific deviation, and τ_f^2 controls **shrinkage strength**. This **borrowes strength** across family members: rare potentials inform from relatives, with **adaptive regularization**—strong shrinkage for similar potentials, weak for diverse families.

2.3.2 Dynamic BMA with Online Model Updating **Sequential Bayesian updating** revises posteriors as new DFT data arrives:

$$p_{t+1}(M_k) \propto p(y_{t+1}|M_k, m_{t+1}) \cdot p_t(M_k)$$

Forgetting factors downweight old data for **non-stationary environments** where potential performance evolves. **Change-point detection** triggers model updates when **prediction errors systematically deteriorate**, indicating concept drift.

2.3.3 Connection to Stacking and Super-Learning Approaches

Method	Weight Learning	Uncertainty	When Preferred
BMA	Marginal likelihood	Full predictive distribution	Model correctness assumed
Stacking	Cross-validation	Bootstrap/approximate	Model misspecification likely
Super-learning	Arbitrary meta-learner	CV-based	Complex interaction patterns
Pseudo-BMA	Stacking weights with BMA form	Hybrid	Practical compromise

Table 10: Comparison of ensemble combination methods, showing complementary strengths.

Stacking often outperforms BMA empirically when the true model is outside the candidate set, but **provides less reliable uncertainty**. **Hybrid approaches** (BMA weights with stacking recalibration) combine **theoretical coherence with empirical robustness**.

3. Acquisition Functions for Potential-Specific Trust Assignment

3.1 Information-Theoretic Acquisition Strategies

3.1.1 Predictive Entropy Search for Minimum Error Location **Predictive Entropy Search (PES)** selects evaluations to **maximize expected reduction in entropy over the optimal potential's location**. For material m and potential k :

$$\alpha_{\text{PES}}(m, k) = H[k^*|D] - \mathbb{E}_{y \sim p(y|m, k, D)}[H[k^*|D \cup \{(m, k, y)\}]]$$

where $k^* = \underset{k}{\operatorname{argmin}} \mathbb{E}[(y_{\text{true}} - \hat{y}_k)^2]$ is the optimal potential. PES **targets decision-relevant uncertainty**: materials where **multiple potentials appear competitive** and **predictions are uncertain** receive highest priority.

Computational challenges: PES requires **integration over posterior distributions**, addressed via **expectation propagation** or **Monte Carlo approximation**. For **GP surrogates**, efficient approximations exploit **analytic predictive forms**. **Batch PES** extends to parallel evaluation, with **greedy selection and submodularity guarantees** ensuring near-optimal diversity.

3.1.2 Max-Value Entropy Search for Best Potential Identification **Max-Value Entropy Search (MES)** targets **reduction in entropy of the maximum achievable accuracy**:

$$\alpha_{\text{MES}}(m, k) = H[y^*|D] - \mathbb{E}_y[H[y^*|D \cup \{(m, k, y)\}]]$$

where $y^* = \min_k \mathbb{E}[(y_{\text{true}} - \hat{y}_k)^2]$. MES is **appropriate when any accurate prediction suffices**, regardless of which potential provides it. This differs from PES: MES values **achieving high performance**; PES values **knowing which potential achieves it**.

Computational advantages: The **maximum value is scalar**, simplifying entropy calculations versus **vector-valued location** in PES. For **23 potentials**, exact MES is **feasible with numerical integration or sampling methods**.

3.1.3 Joint Entropy Reduction Across Material-Property Space **Full uncertainty** in GLIM is **joint across materials, properties, and potentials**. **Joint Entropy Reduction (JER)** targets:

$$\alpha_{\text{JER}}(m, k, p) = H[\mathbf{F}|D] - \mathbb{E}_{y_p}[H[\mathbf{F}|D \cup \{(m, k, y_p)\}]]$$

where $\mathbf{F}$ is the **full performance tensor** and p indexes properties ($\mathbf{K}$, $\mathbf{C}'$, $\mathbf{C}_{44}$).

PCA-based approximation: With **55% variance in PC1**, low-rank factorization $\mathbf{F} = \mathbf{U}\mathbf{V} + \mathbf{E}$ reduces effective dimensionality. JER on **latent factors** $\mathbf{U}$, $\mathbf{V}$ enables **scalable computation**. **Greedy decomposition** via chain rule: $H[\mathbf{F}] = H[f_1] + H[f_2|f_1] + \dots$ prioritizes **most informative materials first**.

3.2 Uncertainty-Aware Acquisition Design

3.2.1 Upper Confidence Bound Formulations with Ensemble Variance Upper Confidence Bound (UCB) balances predicted performance with uncertainty:

$$\alpha_{\text{UCB}}(m, k) = -\mu_k(m) + \beta \sqrt{\sigma_k^2(m) + \sigma_{\text{ensemble}}^2(m)}$$

where $\sigma_{\text{ensemble}}^2(m) = \text{Var}_k[\mu_k(m)]$ is cross-potential variance, capturing epistemic uncertainty about which potential is correct.

Selection	Theoretical Basis	Practical Implementation
$\sqrt{2 \log t}$	UCB1 regret bound	Standard, may over-explore early
$\sqrt{(2 \log(t^{d/2+2})^{2/3})}$	Finite-time guarantee	Confidence parameter
Adaptive: $\alpha_t = \alpha_0/t^\gamma$	Annealing exploration	$\gamma = 0.5$ typical, validated
Empirical Bayes	Maximize held-out reward	Cross-validation on GLIM

Table 11: selection strategies for UCB, balancing theoretical guarantees with practical performance.

Heteroscedastic UCB accommodates input-dependent noise: $\sigma^2(m) = \sigma_{\text{GP}}^2(m) + g(m)$ where $g(m)$ is learned variance function. Materials with high noise but low signal are appropriately deprioritized.

3.2.2 Expected Improvement with Heteroscedastic Noise Models Expected Improvement (EI) quantifies expected reduction in error:

$$\alpha_{\text{EI}}(m, k) = \mathbb{E}[\max(0, f^* - f(m, k)) | D] = \sigma(m)[z\Phi(z) + \phi(z)]$$

where $z = (f(m) - f^*) / \sigma(m)$, Φ and ϕ are standard normal CDF/PDF.

Heteroscedastic extension: $\sigma^2(m) = \sigma_{\text{GP}}^2(m) + \sigma_{\text{noise}}^2(m)$ with learned noise function. Materials with systematically larger errors (complex oxides, high-entropy alloys) receive appropriate uncertainty inflation, preventing overconfident selection.

3.2.3 Knowledge Gradient for Correlated Potential Evaluations Knowledge Gradient (KG) accounts for information propagation through correlations:

$$\alpha_{\text{KG}}(m, k) = \mathbb{E}_{D'}[\max_{k'} \mu_{k'}(m' | D') | D] - \max_{k'} \mu_{k'}(m' | D)$$

where $D' = D \cup \{(m, k, y)\}$. The correlation structure from GLIM PCA—55% variance in PC1—means evaluating one potential informs others. Multi-output GP with coregionalization kernel:

$$K((m, k), (m', k')) = k_m(m, m') \otimes k_k(k, k')$$

enables efficient KG computation via Kronecker structure.

3.3 Multi-Potential Query Strategies

3.3.1 Thompson Sampling for Potential Selection Thompson Sampling implements **probability matching**: sample $\mathbf{k} \sim \mathbf{p}(\cdot | \mathbf{D})$, then select $\mathbf{k} = \arg\max_{\mathbf{k}} \hat{\mu}_{\mathbf{k}}$. This automatically balances exploration and exploitation without tuning: **uncertain potentials have high sampling variance** and are **occasionally selected**; **high-performing potentials have concentrated posteriors** and are **frequently selected**.

Variant	Posterior Representation	Computational Cost	Regret Bound
Standard TS	Exact Gaussian	$O(K^2)$ per decision	$O(\sqrt{KT \log T})$
Linear TS	$\hat{\mu}_{\mathbf{k}} = \mathbf{x}^T \boldsymbol{\theta}_{\mathbf{k}}$	$O(d^2 K)$	$O(d\sqrt{KT})$
GP-TS	Function space sampling	$O(n^3)$ approximate	$O(\sqrt{T} \sqrt{\log T})$
Batch TS	Parallel independent samples	$O(bK^2)$ for batch size b	$O(\sqrt{KT/b})$

Table 12: Thompson Sampling variants for potential selection, with computational and statistical properties.

For GLIM: GP-TS with 23 potentials is tractable with sparse approximations. Batch TS enables parallel DFT evaluation with diversity from sampling stochasticity.

3.3.2 Multi-Armed Bandit Formulations with Correlated Arms Contextual bandits extend to **material-feature-dependent rewards**: $E[r_{\mathbf{k}} | \mathbf{m}] = f_{\mathbf{k}}(\mathbf{m})$ where $\mathbf{m}$ are descriptors. **Linear contextual bandits**: $f_{\mathbf{k}}(\mathbf{m}) = \mathbf{m}^T \boldsymbol{\theta}_{\mathbf{k}}$. **GP contextual bandits**: $f_{\mathbf{k}} \sim \text{GP}(0, \mathbf{k}_{\mathbf{k}})$ with kernel over material space.

Correlation structure from GLIM: 55% PC1 variance suggests low effective dimensionality, enabling efficient learning. Kernelized bandits achieve regret scaling with effective dimension rather than raw material count.

3.3.3 Gittins Indices for Sequential Potential Allocation Gittins indices provide **optimal discounted infinite-horizon policies**. For potential $\mathbf{k}$, the index is **maximum discount rate where $\mathbf{k}$ is preferred to safe alternative**. **Computational challenges**: requires **dynamic programming**, addressed via **Whittle index approximation** or **Monte Carlo tree search**.

Finite-horizon GLIM applications: rollout policies approximate lookahead value, with batch KG providing practical alternative.

4. Active Learning with Model Disagreement Uncertainty

4.1 Query-by-Committee for Interatomic Potentials

4.1.1 Vote Entropy Across 23-Potential Predictions Query-by-Committee (QBC) leverages **ensemble disagreement** to identify informative materials. For discretized predictions into bins b :

$$H_{\text{vote}}(m) = - \sum_b \frac{n_b(m)}{23} \log \frac{n_b(m)}{23}$$

where $n_b(m)$ is the **count of potentials predicting bin b** . **Maximum entropy** (uniform distribution) indicates **strongest disagreement** and **highest information value**.

Disagreement Measure	Formula	Computational Cost	Best For
Vote entropy (binned)	$-\sum_b (n_b/K) \log(n_b/K)$	$O(K)$	Discrete decisions, interpretability
Empirical variance	$(1/(K-1)) \sum_k (\hat{y}_k - \bar{\hat{y}})^2$	$O(K)$	Continuous predictions, simplicity
Average KL divergence	$(1/K^2) \sum_{\{i,j\}} D_{KL}(p_i p_j)$	$O(K^2)$	Probabilistic predictions, full distribution
Mutual information	$I(\cdot; y m)$	$O(K)$ with approximations	Information-theoretic rigor

Table 13: Disagreement measures for QBC, with computational and applicability considerations.

For continuous elastic constants: empirical variance or kernel density-based entropy substitutes for discrete vote entropy. **Materials with $\sigma^2_{\text{ensemble}} > \text{threshold}$ are prioritized for DFT validation.**

4.1.2 Average KL Divergence for Continuous Property Predictions For probabilistic predictions $p_k(y|m)$, the average pairwise KL divergence:

$$D_{KL}(m) = \frac{1}{K^2} \sum_{i,j} D_{KL}(p_i(\cdot|m)||p_{\text{ens}}(\cdot|m))$$

where $p_{\text{ens}} = (1/K) \sum_k p_k$ is the **mixture distribution**. This captures **both mean and variance disagreement**, identifying materials where potentials differ in **confidence as well as point estimates**.

4.1.3 Committee Variance as Proxy for Epistemic Uncertainty Committee variance provides the **simplest, most widely used disagreement measure**:

$$\sigma^2_{\text{committee}}(m) = \frac{1}{K-1} \sum_k (\hat{y}_k(m) - \bar{y}(m))^2$$

Decomposition: Total error² = Bias² + $\sigma^2_{\text{committee}}$ + $\sigma^2_{\text{aleatoric}}$. The **committee variance term** is **reducible epistemic uncertainty**—target of active learning.

Calibration assessment: Correlation of $\sigma^2_{\text{committee}}$ with **held-out DFT error** validates QBC effectiveness. For **well-calibrated ensembles**, **high committee variance predicts high actual error**.

4.2 Uncertainty-Driven DFT Prioritization

4.2.1 Maximum Expected Prediction Error (MEPE) Criteria **Maximum Expected Prediction Error (MEPE)** prioritizes materials with **highest expected squared error**:

$$\alpha_{\text{MEPE}}(m) = \mu^2(m) + \sigma^2(m)$$

combining **systematic bias** () and **uncertainty** (). For **GP surrogates**, this has **closed form** with **heteroscedastic extensions**.

MEPE Component	Source	Reduction Strategy
$\mu^2(m)$	Systematic model bias	Model correction, ensemble centering
$\sigma^2_{\text{GP}}(m)$	GP posterior variance	More data in neighborhood
$\sigma^2_{\text{noise}}(m)$	Irreducible aleatoric noise	Accept or improve measurement precision

Table 14: MEPE decomposition with targeted reduction strategies.

4.2.2 Bayesian Active Learning by Disagreement (BALD) BALD maximizes **mutual information between model parameters and predictions**:

$$\alpha_{\text{BALD}}(m) = H[y|m, D] - \mathbb{E}_{\theta}[H[y|m, \theta]]$$

High BALD: potentials make **confident but different predictions** (high predictive entropy, low expected conditional entropy). **Low BALD**: potentials **agree** or are **individually uncertain**.

4.2.3 Expected Model Change for Potential Improvement Expected Model Change (EMC) selects materials **most altering the model** if known:

$$\alpha_{\text{EMC}}(m) = \mathbb{E}_y[\|\theta - \theta^+(m, y)\|]$$

For **GPs**, **high EMC** corresponds to **high posterior variance**—materials where the **surrogate is most sensitive to new data**.

4.3 Pool-Based and Stream-Based Active Learning

4.3.1 Density-Weighted Uncertainty Sampling Pure uncertainty sampling can target **outliers with low relevance**. Density-weighted uncertainty sampling corrects:

$$\alpha_{\text{DWUS}}(m) = \sigma^2(m) \cdot \hat{p}(m)^\beta$$

Value	Effect	Use Case
$\beta = 0$	Pure uncertainty sampling	Exploration-focused, outlier detection
$\beta = 1$	Uncertainty $\times$ density	Balanced, representative exploration
$\beta > 1$	Strong density emphasis	Conservative, core distribution focus

Table 15 – continued

Value	Effect	Use Case
< 0	Explicit outlier targeting	Robustness testing, anomaly detection

Table 15: Density weighting exponent and its effect on exploration characteristics.

4.3.2 Cluster-Based Representative Selection from JARVIS-FF HDBSCAN clustering in PCA-reduced descriptor space identifies natural material groupings. Per-cluster uncertainty sampling ensures diversity: select most uncertain material from each cluster, proportional to cluster size and aggregate uncertainty.

Hierarchical structure: 2–3 clusters on PC1 (55% variance) for coarse partitioning; finer clustering on higher components for refined diversity.

4.3.3 Adaptive Subsampling for Computational Budget Constraints Dynamic annealing: α decreases as budget depletes, shifting from exploration to exploitation. Budget-aware acquisition: explicit cost integration in value-of-information calculation.

5. Multi-Fidelity and Multi-Objective Bayesian Optimization

5.1 Multi-Fidelity Gaussian Process Frameworks

5.1.1 Linear Autoregressive Coregionalization: Cheap Classical to Expensive ML Potentials
The multi-fidelity framework structures relationships between information sources of varying cost and accuracy. For interatomic potentials, the natural hierarchy is:

Fidelity Level	Methods	Cost (s/atom)	Accuracy	Primary Use
s=1 (Lowest)	EAM, MEAM, Lennard-Jones	~10	Moderate	Large-scale screening, MD
s=2 (Intermediate)	SNAP, GAP, MTP	~10 -10 ¹	Good	Targeted exploration, validation
s=3 (High)	Deep NN potentials (M3GNet, MACE)	~10 ¹ -10 ²	Very good	Accurate property prediction
s=4 (Highest)	DFT, experiment	~10 ² -10	Reference	Ground truth, final validation

Table 16: Multi-fidelity hierarchy for interatomic potential selection, with cost-accuracy trade-offs.

The linear autoregressive model (arXiv.org) :

$$f_s(x) = \rho_{s-1}f_{s-1}(x) + \delta_s(x)$$

where ρ_{s-1} is the correlation coefficient and $\delta_s \sim \text{GP}(0, \mathbf{k}_s)$ is the fidelity-specific residual. This enables information propagation: cheap s=1 evaluations inform expensive s=4 predictions through learned correlations.

Empirical validation: The GP-MFBO framework for temperature/humidity calibration achieved **81.7% improvement in temperature uniformity** and **76.3% in humidity uniformity** over baselines, with **94.2% confidence interval coverage** ([nih.gov](https://www.nih.gov)) . These results **transfer directly** to potential selection: **strategic use of cheap potentials reduces expensive DFT burden dramatically.**

5.1.2 Nonlinear Multi-Fidelity GPs with Deep Kernel Learning Linear coregionalization fails when fidelity relationships are input-dependent. Deep kernel learning addresses this:

$$f_s(x) = \text{GP}(0, k_s(g_s(x), g_s(x')))$$

where g_s is a neural network feature extractor. Learned features identify material subspaces where fidelity correlations are strong, enabling adaptive fidelity selection.

5.1.3 Multi-Output GPs for Correlated Elastic Constants Intrinsic coregionalization model (ICM) for **K, C', C44**:

$$K((m, p), (m', p')) = k_m(m, m') \times B_{p, p'}$$

where **B** is **3×3 coregionalization matrix** learned from GLIM data. **Strong K–C44 correlations in metals, weaker C' correlations in covalent systems** are automatically captured.

5.2 Cost-Aware Acquisition Functions

5.2.1 Expected Improvement per Unit Cost

$$\alpha_{\text{EI/C}}(m, s) = \frac{\mathbb{E}[\max(0, f_s^* - f_s(m))]}{c_s}$$

Normalizes information gain by computational cost, automatically favoring cheap evaluations when informative. For GLIM's 10 cost ratio, EI/C front-loads classical potential screening before selective DFT validation.

5.2.2 Knowledge Gradient with Budget Constraints KGCB optimizes long-term value given total budget **B**:

$$\alpha_{\text{KGCB}}(m, s) = \frac{1}{c_s} \mathbb{E}[\text{Value of information from } (m, s) \text{ for optimal allocation of remaining } B - c_s]$$

Nested optimization is computationally demanding; rollout approximations enable practical implementation.

5.2.3 Look-Ahead Acquisition for Sequential Cost-Benefit Analysis Two-step look-ahead: simulate how cheap evaluation informs subsequent expensive decision. For GLIM: Does EAM evaluation justify ML potential query, or escalate directly to DFT?

5.3 Multi-Objective Pareto Optimization

5.3.1 Expected Hypervolume Improvement for Accuracy-Cost Trade-offs EHVI quantifies expected increase in dominated hypervolume:

$$\alpha_{\text{EHVI}}(m, s) = \mathbb{E}[\text{HV}(P \cup \{(m, s)\}) - \text{HV}(P)]$$

where **P** is current Pareto set and **HV** is hypervolume indicator. For **23 potentials** $\times$ **3 properties** $\times$ **cost**, EHVI identifies evaluations expanding optimal trade-off frontier.

5.3.2 ParEGO for Scalarized Multi-Objective Potential Selection ParEGO randomizes scalarization weights $\sim \text{Dirichlet}()$:

$$f_{\lambda}(m, s) = \lambda_1 \cdot \text{Accuracy}_K + \lambda_2 \cdot \text{Accuracy}_{C'} + \lambda_3 \cdot \text{Accuracy}_{C44} + \lambda_4 \cdot (-\text{Cost})$$

Sequential scalarization with standard **EI** approximates full Pareto frontier without explicit multi-objective optimization.

5.3.3 Multi-Objective Information-Theoretic Acquisition Pareto entropy search targets reduction in entropy of Pareto set location. Computationally demanding but theoretically principled for comprehensive trade-off exploration.

5.4 Adaptive Fidelity Selection

5.4.1 Fidelity-Aware Entropy Search Joint selection of fidelity and location:

$$\alpha_{\text{FES}}(m, s) = \frac{I(\theta; y_{m,s} | D)}{c_s}$$

Information gain per unit cost automatically determines optimal fidelity level.

5.4.2 Multi-Information Source Optimization Non-hierarchical fidelity structure: different potentials excel for different material classes. MISO learns source reliability from historical performance, adaptively selecting best source per query.

5.4.3 Online Fidelity Ranking from GLIM Performance Data Bayesian updates to fidelity correlations: as new (m, s, y) tuples arrive, posterior over $_ \{s, s'\}$ and $_ s$ functions revises. Rapid adaptation to discover unexpectedly informative cheap sources.

6. Batch Active Learning for High-Dimensional Experimental Design

6.1 Batch Mode Uncertainty Quantification

6.1.1 Local Penalization for Diverse Batch Selection Sequential selection with distance penalty:

$$\alpha_{\text{penalized}}(m) = \alpha(m) \cdot \prod_{m_i \in \text{batch}} \phi(\|m - m_i\|)$$

where γ decays with descriptor space distance. Ensures batch diversity without explicit combinatorial optimization.

6.1.2 Determinantal Point Processes for Repulsive Sampling DPP probability: $p(S) \propto \det(K_S)$ where K is similarity kernel. Naturally balances quality (high μ) with diversity (low similarity). Efficient sampling via eigendecomposition.

6.1.3 Thompson Sampling with Correlated Batch Rewards Parallel TS samples: K independent $\hat{\mu}(b) \sim p(\mu | D)$, select $m_b = \arg\max_m \hat{\mu}(b) \cdot f(m)$. Correlation through shared posterior ensures diversity without explicit penalty.

6.2 High-Dimensional Composition and Structure Spaces

6.2.1 Additive Gaussian Processes for Compositional Inputs Decomposition: $f(\text{composition}, \text{structure}) = f_{\text{comp}}(\text{composition}) + f_{\text{struct}}(\text{structure}) + f_{\text{interact}}(\text{composition}, \text{structure})$. Reduces sample complexity and improves interpretability.

6.2.2 Deep Kernel Learning with Structural Descriptors SOAP/ACSF descriptors $\rightarrow$ neural network $\rightarrow$ GP kernel. Learned invariances (rotation, permutation, translation) built into structural fingerprints.

6.2.3 Random Embedding Bayesian Optimization for Latent Spaces Random projection to d_{eff} -dimensional subspace where $d_{\text{eff}} \ll d_{\text{raw}}$. Effective dimensionality from GLIM PCA: 2–3 dimensions capture ~70% variance.

6.3 Parallel and Distributed Acquisition

6.3.1 Constant Liar Strategies for Asynchronous Evaluation Impute pending results with constant value (mean or optimistic). Enables continued selection without waiting for stragglers. “Lie” corrected when true value arrives.

6.3.2 Kriging Believer for Pending Observations Impute with GP posterior mean: more informed than constant liar. Accounts for expected information content of pending evaluations.

6.3.3 Rollout Acquisition for Long-Horizon Planning Simulate future sequences: evaluate value of current decision under optimal continuation. Computationally intensive; limited lookahead (2–3 steps) often sufficient.

7. Gaussian Process Surrogates for Potential Error Prediction

7.1 GP Architectures for Error Surrogates

7.1.1 Standard GP Regression with Matérn and RBF Kernels

Kernel	Smoothness	Computational Cost	Best For
RBF (squared exponential)	Infinitely differentiable	$O(n^3)$	Smooth, well-behaved landscapes
Matérn-3/2	Once differentiable	$O(n^3)$	Moderate irregularity

Table 17 – continued

Kernel	Smoothness	Computational Cost	Best For
Matérn-5/2	Twice differentiable	$O(n^3)$	Recommended default for materials
Rational quadratic	Mixture of length scales	$O(n^3)$	Multi-scale phenomena

Table 17: Kernel selection for GP surrogates in materials applications.

Matérn-5/2 recommended: balance between smoothness and flexibility, avoiding overconfident extrapolation of RBF.

7.1.2 Heteroscedastic GPs for Input-Dependent Noise Twin GP formulation: mean function $\mu(\mathbf{m})$ and log-variance function $\log \sigma^2(\mathbf{m})$ each modeled with independent GPs. Joint optimization via variational inference or expectation propagation.

7.1.3 Warped GPs for Non-Gaussian Error Distributions Parametric warping: $\mathbf{y} \rightarrow \Phi(\mathbf{y})$ where Φ is sinh-arcsinh or similar transformation. Heavy-tailed errors (occasional catastrophic failures) better modeled with appropriate warping.

7.2 Feature Engineering for Material Representation

7.2.1 Compositional Descriptors: Magpie, Oliynyk, and Modified Pettifor Scales

Descriptor Set	Features	Coverage	Best For
Magpie	145 elemental properties + statistics	All inorganic compositions	General-purpose screening
Oliynyk	Ionic radii, electronegativity differences	Ionic/covalent systems	Structure prediction, stability
Modified Pettifor	Single-number chemical scale	Periodic trends	Bonding classification, similarity

Table 18: Compositional descriptor sets for materials representation.

7.2.2 Structural Fingerprints: SOAP, ACSF, and Moment Tensor Descriptors

Fingerprint	Invariance	Completeness	Computational Cost
SOAP	Rotation, translation, permutation, reflection	Complete basis	Moderate ($O(n_{\text{atoms}} \times n_{\text{species}})$)
ACSF	Same as SOAP	Hand-designed, incomplete	Fast (precomputed functions)
Moment tensors	Rotation, translation, permutation	Polynomial-complete	Moderate

Table 19: Structural fingerprints for local atomic environment description.

7.2.3 Learned Representations from Crystal Graph Convolutional Networks CGCNN, MEGNet, ALIGNN: end-to-end learned embeddings from crystal graphs. Pre-trained on large databases, fine-tuned on GLIM benchmark. Task-specific representations may outperform hand-engineered features.

7.3 Uncertainty Calibration and Validation

7.3.1 Probability Integral Transform for Calibration Assessment PIT histogram: uniform for well-calibrated predictions. U-shaped: underconfidence; inverse-U: overconfidence. Recalibration via temperature scaling or isotonic regression.

7.3.2 Conformal Prediction for Distribution-Free Uncertainty Coverage guarantee: $P(y_{\text{true}} \in C_{\alpha}(x)) \geq 1 - \alpha$ for any distribution, any model. Split conformal: efficient, valid with exchangeability.

7.3.3 Cross-Validation Strategies for Large-Scale Benchmarks

CV Strategy	Structure	Use Case
Random	Shuffle all data	Optimistic, assesses interpolation
Leave-one-material-out	Hold out single material	Realistic generalization
Leave-one-cluster-out	Hold out material cluster	Extrapolation to new classes
Time-based	Train on older, test on newer	Temporal generalization

Table 20: Cross-validation strategies for GLIM-scale benchmarks.

7.4 Scalable GP Approximations

7.4.1 Sparse Variational GPs with Inducing Points Variational lower bound: $O(nm^2)$ for $m \ll n$ inducing points. Optimization of inducing locations and kernel parameters jointly. Modern implementations: millions of data points.

7.4.2 Random Fourier Features for Kernel Approximation Explicit feature map: $k(x, x') \approx \frac{1}{d} \sum_{i=1}^d \phi_i(x) \phi_i(x')$ with $\phi_i(x) = \cos(w_i x + b_i)$. Primal space regression: $O(nd^2)$ for d features, efficient when $d \ll n$.

7.4.3 Neural Network-Based Mean Functions for Complex Error Landscapes Deep network mean + GP residual uncertainty: $f(x) = \text{NN}(x) + \sigma(x)$ where $\sigma \sim \text{GP}(0, k)$. Combines representation learning with calibrated uncertainty.

8. Applicability Domain Methods from QSAR/Cheminformatics

8.1 Distance-Based Domain Characterization

8.1.1 Leverage Statistics in Descriptor Space Leverage: $h_i = x_i (X^T X)^{-1} x_i$. High h_i : extrapolative relative to training data. Warning threshold: $h_i > 3p/n$ typically.

8.1.2 Hotelling’s T² for Multivariate Outlier Detection $T^2 = (x - \bar{x})^T S^{-1} (x - \bar{x})$. Accounts for feature correlations. Extreme values: underrepresented regions where predictions unreliable.

8.1.3 k-Nearest Neighbor Distance in Feature Space Nonparametric density proxy: large distance to k-th nearest neighbor indicates low training density. Adaptive k based on local density variation.

8.2 Density-Based and Probabilistic Approaches

8.2.1 Kernel Density Estimation for Training Set Distribution KDE: $\hat{p}(x) = (1/n) \sum_i K_h(x-x_i)$. Bandwidth h: cross-validated or rule-of-thumb. Low $\hat{p}(x)$: domain violation.

8.2.2 One-Class SVM for Novelty Detection Decision boundary enclosing training data. Kernel and regularization control boundary tightness. Trade-off: false positives vs. false negatives.

8.2.3 Gaussian Mixture Models for Multi-Modal Domains Multi-modal training distribution: materials cluster into distinct classes (metals, oxides, semiconductors). GMM captures this structure, flagging low-probability regions.

8.3 Ensemble-Based Domain Definitions

8.3.1 Standard Deviation of Ensemble Predictions $\sigma_{\text{ensemble}}(x)$: natural applicability domain measure. No explicit descriptor construction required. High : ensemble uncertain, domain boundary likely.

8.3.2 Bootstrap Aggregation Confidence Intervals Empirical distribution across bootstrap samples. Narrow intervals: reliable predictions; wide intervals: uncertain, near boundary.

8.3.3 Conformal Prediction Sets for Coverage Guarantees Explicit error rate control: $(1-\alpha)$ coverage guaranteed. For potential selection: rigorous statements about trustworthiness.

8.4 Adaptation to Materials-Specific Challenges

8.4.1 Composition-Structure-Property Relationship Modeling Non-unique mapping: similar compositions $\rightarrow$ different structures $\rightarrow$ different properties. Joint modeling of composition $\rightarrow$ structure $\rightarrow$ property required for reliable domain definition.

8.4.2 Transferability Across Crystal Systems and Chemistries Critical challenge: potential trained on fcc metals fails for hcp or bcc. Explicit crystal system indicators or learned representations must detect such boundaries.

8.4.3 Dynamic Domain Updating with Active Learning Conservative initial domain, relaxed as confidence grows. Explicit tracking of domain expansion vs. exploitation of known regions.

9. GLIM-Style Meta-Analysis: Practical Implementation

9.1 Exploratory Data Analysis of Potential Performance

9.1.1 Performance Clustering via PCA and t-SNE PCA: 55% variance in PC1 $\rightarrow$ global quality axis. t-SNE/UMAP: nonlinear clustering revealing fine-grained structure. Materials with similar potential rankings cluster together $\rightarrow$ transfer of selection decisions.

9.1.2 Correlation Structure Across Elastic Constants

Property Pair	Typical Correlation	Bonding Type Dependence
K-C44	Strong positive	Metallic systems
K-C'	Moderate	Variable by structure
C' -C44	Weak to moderate	Strongest in covalent systems

Table 21: Elastic constant correlations and their bonding-type dependence.

9.1.3 Systematic Error Patterns by Material Class Regression analysis: potential error $\sim$ material class indicators. Quantified patterns: EAM excellent for simple metals, fails for oxides; MLIAPs more consistent but slower.

9.2 Meta-Learning for Potential Recommendation

9.2.1 Learning to Learn: Feature-Based Meta-Models Meta-model: predict best potential from material descriptors. Training: GLIM performance matrix. Zero-shot generalization to new materials.

9.2.2 Prototypical Networks for Material-Potential Matching Prototype per potential: mean embedding of materials where it excels. New material: classify to nearest prototype. Interpretable, uncertainty via distance.

9.2.3 Graph Neural Networks for Relational Potential Ranking Bipartite graph: materials potentials with performance edges. Message passing: embed materials and potentials jointly. Link prediction: unobserved performance estimation.

9.3 Online Decision Frameworks

9.3.1 Contextual Bandits for Material-Adaptive Selection Linear/Neural contextual bandits: $\text{reward}(\text{potential}|\text{material features})$ learned online. Regret bounds: sublinear growth, asymptotic optimality.

9.3.2 Bayesian Optimization with Meta-Priors from GLIM Strong prior from benchmark: rapid convergence for new materials. Warm start: posterior concentrated near GLIM-informed values.

9.3.3 Regret Bounds and Asymptotic Optimality Guarantees Theoretical justification: GP-UCB, Thompson Sampling achieve $O(\sqrt{(\log T) T})$ regret. $\log T$: maximum information gain, depends on kernel and dimensionality.

10. Integration and Practical Implementation Guidance

10.1 Software Ecosystem and Tools

Tool	Purpose	Key Features	Integration
BoTorch/GPyTorch	Modern Bayesian optimization	GPU acceleration, auto-diff, modular acquisitions	Primary framework
scikit-learn/GPy	Classical GP implementations	Accessible, stable, well-documented	Prototyping, education
ASE	Atomistic simulation	Unified calculator interface, MD engines	Potential evaluation
pymatgen	Structure manipulation	Crystal analysis, I/O, builders	Descriptor computation
Matminer	Feature extraction	Magpie, SOAP, structure featurizers	Pipeline integration
AiiDA/FireWorks	Workflow orchestration	Provenance, distributed execution	Scale-out deployment

Table 22: Software ecosystem for Bayesian optimization of interatomic potential selection.

10.2 End-to-End Workflow Design

10.2.1 Offline Preprocessing: GLIM Benchmark Ingestion and Feature Computation One-time cost: descriptor computation, similarity matrices, meta-model training. Cached representations accelerate online decisions.

10.2.2 Online Decision Loop: Query, Evaluate, Update, Recommend

```

Input: Material m with descriptors  $\phi(m)$ 
1. Predict:  $\mu_k(m), \sigma_k(m)$  for all k potentials via GP surrogate
2. Select:  $(m, k^*) = \operatorname{argmax} \operatorname{acquisition}(\mu, \sigma, \text{cost})$ 
3. Evaluate: Run potential  $k^*$  on material m, observe y
4. Update: Incorporate  $(m, k^*, y)$  into GP posterior
5. Recommend: Return best predicted potential or BMA prediction

```

10.2.3 Human-in-the-Loop Validation for Critical Applications Expert override: uncertainty estimates, similar materials, known failure modes presented for review. Feedback improves future automation.

10.3 Validation and Benchmarking Protocols

10.3.1 Leave-One-Material-Out Cross-Validation Primary metric: generalization to new materials. Correlation of predicted vs. actual best potential.

10.3.2 Time-Based Splits for Temporal Generalization Training on older benchmark versions, testing on newer. Detects obsolescence as potentials improve.

10.3.3 Adversarial Testing for Robustness Assessment Deliberately challenging materials: extreme compositions, unusual structures, engineered outliers. Stress testing reveals failure modes.

10.4 Future Directions and Open Challenges

10.4.1 Foundation Models for Universal Potential Pre-Training Large-scale pre-training on diverse materials data → strong priors reducing need for material-specific selection. Uncertainty quantification and domain adaptation remain critical.

10.4.2 Causal Inference for Mechanistic Understanding Beyond correlation: identify why potentials fail. Structural causal models guide targeted improvement rather than black-box selection.

10.4.3 Federated Learning for Distributed Potential Development Multi-institutional collaboration without centralizing proprietary data. Bayesian methods naturally handle heterogeneous, incomplete information.